Abhishek Sampath Kumar

Postdoctoral Fellow · **Denise A. and Eugene W. Chinery Research Scholar** | Stem Cell and Regenerative Biology, Harvard University | Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard

Cambridge, MA · abhisheksampathkumar@fas.harvard.edu · abhisampathkumar.github.io

Google Scholar · ORCID · NCBI bibliography

POSITIONS

2023-present **Postdoctoral fellow**, Stem Cell and Regenerative Biology, Harvard University · advisor Paola Arlotta

2023-present **Research affiliate**, Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard

2018–2023 **Graduate student**, Max Planck Institute for Molecular Genetics, Berlin · advisor Alexander Meissner

2017 **Research fellow**, Max Planck Institute for Molecular Genetics, Berlin · advisor Edda Schulz · master's thesis research

2016 **Research assistant**, Institute for Botany and Plant Physiology, University of Jena

2013–2015 **Research fellow**, Institute of Molecular and Cell Biology, A*STAR, Singapore · advisor Daniel Messerschmidt

2013 **Research fellow**, Cancer Science Institute, National University of Singapore · advisor Phillip Koeffler · undergraduate thesis research

EDUCATION

2023 **PhD, Molecular Genetics** · Technische Universität Berlin
Research at the Max Planck Institute for Molecular Genetics
Thesis: Spatiotemporal architecture of mouse embryogenesis at single-cell resolution

2017 **MSc, Molecular Life Sciences** · Friedrich Schiller University, Jena
Master's thesis: Investigation of putative regulators of X-chromosome inactivation at single-cell resolution

2013 **BTech, Genetic Engineering** · SRM University, Chennai
Bachelor's thesis: Identification and characterization of somatic mutations in acute myeloid leukemia

AWARDS AND FELLOWSHIPS

2026-present **Denise A. and Eugene W. Chinery Research Scholar**

2025 **The John Hatch Trainee Leadership Award**, Stem Cell and Regenerative Biology, Harvard University

2025 **Postdoc Professional Development Travel Award**, Stem Cell and Regenerative Biology, Harvard University

2025 **Bauer Discovery Grant**, 10x Genomics, Oxford Nanopore and the Harvard Bauer Core

2025 **Runner-up**, PARSE Neuroscience Grant, PARSE Bioscience

2013 **Merit scholarship**, SRM University, Chennai

PUBLICATIONS

Selected publications; the complete list with research highlights sits on the publications page.

LEAD AUTHOR

1. Spatiotemporal transcriptomic maps of whole mouse embryos at the onset of organogenesis · ***Nature Genetics*** 55, 1176–1185 (2023) · Paper

2. Hijacking of transcriptional condensates by endogenous retroviruses · ***Nature Genetics*** 54, 1238–1247 (2022) · Paper

3. Loss of maternal Trim28 causes male-predominant early embryonic lethality · ***Genes & Development*** 31, 12–17 (2017) · Paper

CO-AUTHOR

4. Human brain organoids record the passage of time over multiple years · ***Nature*** (2026) · Paper

5. Organoids of the human ganglionic eminence reveal biased development of medial over caudal ganglionic eminence lineages in autism risk-gene mutants · in revision, *Nature Neuroscience* (2026)

6. Spatially resolved single-nucleus DNA methylation and transcriptome sequencing · in revision, *Cell* (2026)

7. Biobanking pre-patterned cortical organoid progenitors for scalable brain chimeroid production · in revision, *Nature Protocols* (2026)

8. Brain Chimeroids reveal individual susceptibility to neurotoxic triggers · ***Nature*** 631, 142–149 (2024) · Paper

9. Systematic comparison of sequencing-based spatial transcriptomic methods · ***Nature Methods*** 21, 1743–1754 (2024) · Paper

10. TET activity safeguards pluripotency throughout embryonic dormancy · ***Nature Structural & Molecular Biology*** 31, 1625–1639 (2024) · Paper

11. Basal delamination during mouse gastrulation primes pluripotent cells for differentiation · ***Developmental Cell*** 59, 1252–1268 (2024) · Paper

12. GATA transcription factors drive initial Xist upregulation after fertilization · ***Nature Cell Biology*** 25, 1704–1715 (2023) · Paper

13. Dynamic antagonism between key repressive pathways maintains the placental epigenome · ***Nature Cell Biology*** 25, 579–591 (2023) · Paper

14. Emergence and patterning dynamics of mouse-definitive endoderm · ***iScience*** 25, 103556 (2022) · Paper

15. Dnmt1 has de novo activity targeted to transposable elements · ***Nature Structural & Molecular Biology*** 28, 594–603 (2021) · Paper

16. Epigenetic regulator function through mouse gastrulation · ***Nature*** 584, 102–108 (2020) · Paper

17. Mouse embryonic stem cells self-organize into trunk-like structures with neural tube and somites · ***Science*** 370, eaba4937 (2020) · Paper

18. Profiling of somatic mutations in acute myeloid leukemia with FLT3-ITD at diagnosis and relapse · ***Blood*** 126, 2491–2501 (2015) · Paper

19. Laminin-5γ-2 is highly expressed in anaplastic thyroid carcinoma · ***Journal of Clinical Endocrinology and Metabolism*** 99, E62–72 (2014) · Paper

INVITED TALKS

2025 *Maintenance of lifelong neuronal subtype identity* · Cell Circuits and Epigenomics Seminar Series, Broad Institute of MIT and Harvard

2023 *Spatiotemporal architecture of mouse embryogenesis at single-cell resolution* · Takara Bio spatial genomics webinar series · Watch

2021 *Hijacking of transcriptional condensates by endogenous retroviruses* · Cell Circuits and Epigenomics Seminar Series, Broad Institute of MIT and Harvard

TEACHING AND MENTORING

2026–2027 **Research mentor**, Ila Camara, Harvard College · undergraduate research thesis

2026–2027 **Research mentor**, Maria Chifor, Heidelberg University

2025 **Teaching fellow**, SCRB135 Reproductive Biology, Harvard University · instructor Kara McKinley

2024–2025 **Research mentor**, Gabriel Barakat, Boehringer Ingelheim Research Fellow · project design and data analysis

2024 **Teaching fellow**, SCRB195 The Translational Science of Stem Cells, Harvard University · instructor Lee Rubin

LEADERSHIP AND SERVICE

2024–present **Founding organizing member**, Slice of Science, a discussion series promoting cross-disciplinary dialogue and collaboration · Stem Cell and Regenerative Biology, Harvard University

2024–present **Organizing committee**, Postdoc Professional Development Committee, working on research culture, career development and community among postdocs · Stem Cell and Regenerative Biology, Harvard University

2024–present **Communications chair**, Harvard Faculty of Arts and Sciences Postdoctoral Association

Peer reviewer for ***BMC Genomics***, ***Scientific Reports*** and ***STAR Protocols***

2011–2013 **Secretary**, Students' Association, Department of Genetic Engineering, SRM University

OUTREACH

2024–present **Speaker and volunteer**, Meet a Scientist, Broad Institute of MIT and Harvard · stem cell and neuroscience research for public audiences

2018–2022 **Speaker and volunteer**, Long Night of the Sciences, Max Planck Society, Berlin · developmental and regenerative biology for non-specialist audiences